

PVC SUSPENSION REACTOR VER 1

Operator Training Simulator · batch suspension-PVC reactor R-101 (140 m³)

Chulalongkorn University · Department of Chemical Engineering

DOCUMENT NO. : PVC-V1-IN-CE-001

DOCUMENT TITLE : CAUSE AND EFFECT DIAGRAM

DRAWINGS : PVC-PID-001 / PVC-PID-002

Teaching document generated from the simulator logic (bmsTrip, bmsStep, tripConditions, bmsCmd permissives, alarm list).

Rev.	Date	Revision status	Prepared	Checked	Approved	Remark
R0	06-Oct-26	Issued for teaching	A. Soottitantawat	—	—	

PVC SUSPENSION REACTOR VER 1 · R-101 CAUSE and EFFECT DIAGRAM Chulalongkorn University · Department of Chemical Engineering								Client :		Chulalongkorn University · Department of Chemical Engineering								REV	DATE		ORIGINATOR		CHECKER		APPROVER	
								PROJECT :		PVC Suspension Reactor Ver 1 · OTS								R0	06-Oct-26		A. Soottitantawat		—		—	
								DOCUMENT No. :		PVC-V1-IN-CE-001																
								DOCUMENT TITLE :		Cause and Effect Diagram																
								P&ID :		PVC-PID-001 / PVC-PID-002															SHEET 1 of 1	
Causes								EFFECTS														Notes: 1. Action codes: O = open; C = close (FIC-103: to MAN, 0 %); FO = forced fully open; TP = tripped and latched (manual reset); F = first-out; A = activate; P = permissive (the valve cannot be opened until the condition is met); RB = reset blocked. 2. The short-stop XV-104 opens only when the initiator is in and no short-stop has been injected; before the reaction a trip closes the charges only. 3. After the short-stop the SIS forces full cooling while it stays tripped: TV-110A (steam) shut, TV-110B, FV-120 and FV-130 at 100 % whatever their controllers ask. 4. ESD push-button HS-001: immediate, no delay; trips as items 002-005 and also opens the emergency vent XV-108 to D-103 (closed again on reset). 5. XS-101 and FSLL-110 are armed only while the batch is reacting (initiator in, not short-stopped, free monomer left); XS-101 reads the motor-run contact and does not see a broken coupling. 6. LSHH-100 is not latched and gives no short-stop: each time the level passes 94 % for 5 s it shuts XV-101, XV-102 and puts FIC-103 in MAN 0 %. A passing FV-103 keeps injecting (no block valve, HAZOP R17). 7. Time delays are applied in the logic solver. Reset at the SIS console once TSHH-101, PSHH-101 and LSHH-100 are clear; the charge valves stay shut until restarted from the batch sequence. 8. Process alarms are suppressed by group: grade band, level, baffle CW and injection while not reacting; jacket flow while R-101 is empty; agitator power while M-101 is off or the level is below 40 %; E-101 while the condenser carries < 150 kW. 9. PSV-100 (15.0 bar g) relieves to D-103; it is a relief device, shown for completeness of the R-101 annunciation, not an SIS cause. 10. Sequence permissive, not a trip: no valve moves; the operator cannot open the valve or reset until the condition clears. 11. FSLL-110 uses FT-110, the same transmitter as the FI-110 alarm and the batch permissive; LSHH-100 is drawn on LT-100 on PVC-PID-001 (the simulator trips on its own level measurement, see the P&ID tab, section 5). 12. Instrument-air failure: on/off valves fail closed and control valves go to their fail positions; PVC-PID-001 draws XV-104 and XV-108 fail-closed (HAZOP R39: air receiver).				
								XV-104 · Short-stop injection valve from pot D-105	XV-101 · Water charge valve (fail closed)	XV-102 · VCM charge valve (fail closed)	XV-103 · Initiator dosing valve	XV-109 · Vacuum valve to VP-101	XV-106 · Bottom discharge valve to D-102	FIC-103 · Injection water FV-103 to MAN 0 %	TV-110A · LP steam to the jacket loop	TV-110B · CW injection to the jacket loop	FV-120 · Baffle cooling water	FV-130 · E-101 condenser cooling water	XV-108 · Emergency vent to dump tank D-103	SIS · SIS trip latch (manual reset)	1ST · First-out annunciation					RED · SIS alarm on the DCS (red)
Trip Action								O	C	C	P	C	P	C	C	FO	FO	FO	O	TP	F	A	A	A		
Item No.	Tag No.	Tag Service Description	Trip/Alarm-Direction	Setting	Eng. Unit	Delay	Note no.																			
001	HS-001	Emergency shutdown push-button (Batch & SIS tab)	Pushed	—	—	0 s	4	X	X	X		X		X	X	X	X	X	X	TP	X	X		X		
002	TSHH-101	R-101 temperature high-high on TE-101 (independent of TE-100)	High-High	63	°C	10 s	2, 3	X	X	X		X		X	X	X	X	X		TP	X	X		X		
003	PSHH-101	R-101 pressure high-high on PT-101 (independent of PT-100)	High-High	12.5	bar g	5 s	2, 3	X	X	X		X		X	X	X	X	X		TP	X	X		X		
004	XS-101	Agitator M-101 stopped while reacting (motor-run contact)	Stopped	—	—	15 s	2, 3, 5	X	X	X		X		X	X	X	X	X		TP	X	X		X		
005	FSLL-110	Jacket circulation low-low while reacting (FT-110)	Low-Low	120	m³/h	30 s	2, 3, 5, 11	X	X	X		X		X	X	X	X	X		TP	X	X		X		
006	LSHH-100	R-101 level high-high: charges and injection stopped (not latched)	High-High	94	%	5 s	6, 11		X	X				X								X		X		
007	TIC-100	R-101 temperature low (grade band, TE-100)	Low	56.0	°C		6																X	X		
008	TIC-100	R-101 temperature high (grade band, TE-100)	High	58.0	°C		6																X	X		
009	TI-101	R-101 temperature high on TE-101 (pre-trip)	High	60.0	°C																		X	X		
010	PI-100	R-101 pressure high on PT-100 (pre-trip)	High	11.0	bar g																		X	X		
011	LIC-100	R-101 level low (LT-100)	Low	74	%		6																X	X		
012	LIC-100	R-101 level high (LT-100)	High	88	%		6																X	X		
013	FI-110	Jacket circulation low (FT-110)	Low	300	m³/h		6																X	X		
014	JI-100	Agitator M-101 power low	Low	150	kW		6																X	X		
015	JI-100	Agitator M-101 power high (lumps)	High	420	kW		6																X	X		
016	FIC-120	Baffle cooling water low	Low	15	m³/h		6																X	X		
017	TI-131	E-101 cooling-water return hot (scaling)	High	58	°C		6																X	X		
018	TI-132	E-101 top cold: inerts blanketing the condenser	Low	46	°C		6																X	X		
019	FIC-103	Injection water high	High	8	m³/h		6																X	X		
020	TI-120	Cooling-water supply warm	High	35	°C																		X	X		
021	TIC-110	Jacket inlet above 75 °C with the initiator in (first hour)	High	75	°C																		X	X		
022	PD-100	End point: PI-100 has fallen 1.8 bar below its running maximum	Reached	1.8	bar																		X	X		
023	XA-133	Foam / PVC carried into E-101	Detected	—	—																		X	X		
024	XV-105	Degassing valve opened on a batch that is still reacting	Open	—	—																		X	X		
025	PSV-100	Relief valve lifting to dump tank D-103	Lifting	15.0	bar g		9																X	X		
026	SIS-RST	SIS reset permissive: TSHH-101, PSHH-101 and LSHH-100 clear	Not met	63 / 12.5 / 94	°C / bar g / %		7, 10													RB						
027	PERM-VCM	VCM charge permissive: SIS healthy, XV-106 shut, evacuated, water ≥ 95 %, M-101 running, FI-110 ≥ 300 m³/h, P-103 running, XV-109 shut	Not met	—	—		10			P																
028	PERM-INIT	Initiator permissive: as VCM charge, plus VCM charged ≥ 95 % and TI-100 ≥ 54 °C	Not met	54	°C		10				P															
029	PERM-EVAC	Vacuum permissive: SIS healthy, XV-106 shut, PI-100 ≤ 1.0 bar g	Not met	1.0	bar g		10				P															
030	PERM-DISCH	Discharge permissive: PI-100 ≤ 2.0 bar g, M-101 running, charge valves shut	Not met	2.0	bar g		10					P														
031																										
032																										